

Aryan Sarswat

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EXPERIENCE

Machine Learning Scientist II

Jul 2025 – Present

Expedia Group

Seattle, WA

- **Agentic Evaluation Platform:** Designed evaluation framework for multi-turn **ReAct** agents combining deterministic tool-call metrics (correctness, argument validity) with **LLM-as-judge** scoring across tone, responsiveness, efficiency, and grounding; built persona-conditioned **traveler LLM simulator** enabling multi-turn rollout evaluation beyond single-turn benchmarks; shipped annotation dashboard and pipeline accelerating golden dataset curation for faster eval iteration
- **MCP Evaluation Framework:** Architected multi-stage extendable evaluation pipeline for tool call classification, argument validity, hallucination detection, and instruction following; built system to reverse-engineer user prompts from production Splunk logs (MCP tool arguments) to generate **1K+ sample synthetic dataset** for regression testing without access to raw OpenAI queries; designed analysis dashboard adopted as **company-wide AI quality framework** across multiple teams
- **Eval Coverage Analyzer:** Built tool measuring whether golden eval datasets represent production traffic by reframing eval quality as **embedding-space coverage**; reports nearest-neighbor recall@k, applies LLM-aided clustering, and computes **Jensen-Shannon divergence** between production and eval distributions to quantify gaps and drift
- **Expedia Group Travel MCP:** Core contributor to Expedia's MCP server now powering third-party agentic experiences on **Claude and ChatGPT**; owned **33+ PRs** across tool development, schema design, and compliance, iterating prompts to reach **100% tool selection accuracy** and **86% argument validity** with safety guardrails; product peaked at **15K users** within 2-week OpenAI partnership launch
- **PromptLab:** Developed internal agentic prompt engineering platform enabling technical & non-technical users to iterate on prompts with auto-generated test cases, evaluation-driven improvements, and multi-LLM provider support
- **PII Redaction Pipeline:** Fine-tuned open-source models for PII detection and redaction, delivering **~82% F1 uplift** over legacy CRF baseline; deployed to production for safer downstream LLM consumption
- **Hackathon Semi-Finals:** Led team to semi-finals building AI filter prototype that uses **traveler memories** to surface personalized property filters; concept later shipped to production with real-time user queries as the signal
- **Flight Delay Prediction:** Built end-to-end CatBoost system on **78M+ rows** reaching **89.9% accuracy** and operationalized FastAPI real-time inference impacting **38M+ passengers**

Machine Learning Science Intern

May 2024 – Jul 2024

Expedia Group

Seattle, WA

- Developed **Image2Text pipeline** using VLMs (PaliGemma, BLIP-2) with quantized Llama3; deployed to production for image ranking, generating **\$5.4M GP** and **\$1.85M travel ad spend uplift**
- Built Image2Video pipeline (DynamiCrafter, Swin2SR, FLAVR) and Image Relevancy pipeline (CLIP, BLIP) with custom search for destination imagery matching

Research Assistant

Aug 2022 – July 2023

*Agency for Science, Technology and Research (A*STAR)*

Singapore

- Developed state-of-the-art deepfake detection models achieving **10% improvement** over existing methods on novel datasets; formulated scalable video processing pipeline handling datasets up to **1TB**

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Master of Science in Computer Science; GPA: 4.0/4.0

Aug 2023 – May 2025

- Concentration: Artificial Intelligence, Perception, and Natural Systems
- Coursework: Deep Learning, NLP, Reinforcement Learning, Computer Vision, Efficient ML

National University of Singapore

Singapore

Bachelor of Computing with Distinction; Specialization in Artificial Intelligence

Aug 2019 – May 2023

- University Scholar Programme (USP): Multidisciplinary honors program for top 1% of students

SELECTED PROJECTS

- Cobalt: Scalable Teleoperation for Robot Learning** | *Masters Thesis* Jan 2024 – May 2025
- Developed teleoperation platform to democratize robot learning at scale in simulation and real world
 - Leveraged vectorized environments with load-balanced infrastructure supporting concurrent multi-user teleoperation on single GPU
- Efficient Backdoor Unlearning in Language Models** Jan 2025 – May 2025
- Explored early-stage detection and removal of backdoors in LMs without retraining using SPECTRE’s robust covariance estimation adapted to language modeling context
 - Performed gradient ascent on detected poisoned samples; demonstrated early intervention reduces ASR while maintaining model accuracy with significantly lower computational overhead
- Efficient-Selection-via-Pruning (ESVP)** Jan 2024 – May 2025
- Novel data selection approach using model pruning and quantization for active learning; achieved **1.2-2x speed-up** with structured pruning, **1.35x** with quantization, learning with **50% less data** at only 4% F1 loss
- Zero-Shot Adaptation for Guided Robot Policies** Aug 2024 - Dec 2024
- Introduced framework combining diffusion-based trajectory planners with LLM-generated loss functions for zero-shot policy adaptation; achieved real-time adaptation to spatial and speed constraints in MetaWorld
- Historical Document Restoration with GANs** Jan 2025 – May 2025
- Trained Pix2Pix GAN to reconstruct damaged documents; fine-tuned TrOCR for baselines; demonstrated GAN-based restoration significantly improves OCR accuracy on simulated degradations
- Deep RL in 3D Connect 4** Jan 2024 – May 2024
- Investigated DQN, Actor-Critic, and PPO via self-play; achieved 10-20% success rate against Minimax depth-2, consistently outperforming random and shallow Minimax agents

HONORS & AWARDS

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| Hackathon Semi-Finalist – Expedia Group | 2025 |
| Hackathon Microsoft, GenAI, and Assurant Challenge Winner – Georgia Institute of Technology | 2024 |
| Certificate of Distinction in AI – National University of Singapore | 2022 |
| USP Senior Honour Roll – National University of Singapore | 2021 |